ANUTHTHARA RUPASINGHE

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SUMMARY

Researcher and educator with a Ph.D. in Electrical Engineering and postdoctoral research experience at the Princeton Neuroscience Institute. Expertise in statistical machine learning, Bayesian inference, signal processing, and data analysis, with applications to neural, physiological, and behavioral data. Experienced in university-level teaching, including independent course instruction, with a strong interest in teaching and mentoring students in mathematics, engineering, and computational methods.

EDUCATION

Ph.D., Electrical and Computer Engineering *2017 - 2022*
University of Maryland, College Park Cumulative GPA: 3.97/4.00

B.S. (Honors), Electrical and Electronic Engineering *2013 - 2016*
University of Peradeniya, Sri Lanka Cumulative GPA: 4.00/4.00

SKILLS

Programming:	Python (PyTorch, NumPy, Pandas, SciPy), MATLAB
Data:	neuronal spiking data, calcium imaging, EEG, EMG, high-dimensional time series data
Statistical Modeling:	Bayesian inference, Gaussian processes, generalized linear models, logistic regression
AI / ML:	Probabilistic modeling, predictive modeling, regression, clustering, model validation
Methods:	Variational inference, optimization, uncertainty quantification, variance modeling, calibration
Signal Processing:	Spectral analysis, Kalman filtering, frequency-domain inference, causal inference
Research Skills:	Data analysis, model evaluation, technical documentation, interdisciplinary collaboration
Tools:	Git, LaTeX, Jupyter, MS Office Package

RESEARCH EXPERIENCE

Princeton Neuroscience Institute, Princeton University 2022 - present
Postdoctoral Research Fellow, Advisor: Dr. Jonathan Pillow

- Developing statistical and machine learning models for neural data analysis, with emphasis on neural decoding, interpretable probabilistic modeling, uncertainty quantification, and continuous behavioral prediction.
- Developed **Continuous Multinomial Logistic Regression**, a Bayesian machine learning framework for decoding continuous behavioral variables from high-dimensional neural population activity, producing full conditional distributions rather than only point estimates. [Python]
- Implemented scalable inference methods for CMLR using Gaussian process priors, frequency-domain computation, and stochastic variational inference for high-dimensional neural decoding.
- Built analysis pipelines to train, evaluate, and validate decoding models across large-scale datasets from visual cortex, hippocampus, and motor cortex, including assessments of predictive accuracy, posterior uncertainty, calibration, and model robustness.
- Developed **Continuous Modulated Poisson** modeling, a continuous-time statistical framework for modeling over-dispersed neuronal spiking activity and separating stimulus-driven responses from latent modulatory variability. [MATLAB]
- Designed scalable frequency-domain variational inference and validation pipelines to compare model likelihoods, estimate variability structure, and interpret neural population dynamics across sensory areas.
- Authored manuscripts, technical descriptions, and analysis documentation for neuroscience and machine learning audiences, including work accepted at ICLR and eLife.
- Collaborated with interdisciplinary researchers to translate statistical modeling results into interpretable scientific conclusions.

University of Maryland, College Park 2018 - 2022
Graduate Research Assistant, Advisor: Dr. Behtash Babadi

- Developed Bayesian statistical models and inference algorithms for high-dimensional neural and physiological time-series data, with applications to neuronal spiking activity, calcium imaging, EEG, EMG, and human heart rate variability.
- Developed a Bayesian multitaper spectral estimation methodology for multivariate neuronal spiking time-series data, combining Kalman filtering, frequency-domain inference, generalized linear models, and state-space modeling. [MATLAB]
- Developed statistical models and algorithms for adaptive frequency-domain Granger causal inference from multivariate neuronal spiking data, combining Kalman filtering, generalized linear models, and state-space modeling.
- Mentored a graduate student on Granger causal inference from spiking observations via latent variable modeling.
- Developed Bayesian and variational inference methods to directly estimate signal and noise correlations from two-photon calcium imaging data without intermediate spike deconvolution, in collaboration with the Kanold Lab at Johns Hopkins University. The model combines point-process modeling, state-space methods, Pólya-Gamma augmentation, and variational inference to relate latent spiking activity directly to observed fluorescence traces. [MATLAB]
- Analyzed whole-brain light-sheet imaging, EEG, and EMG data from larval zebrafish using spectrotemporal analysis and machine learning techniques to study respiration-locomotion coordination, in collaboration with the Ahrens Lab at Janelia Research Campus, HHMI. [Python]

· Extended multitaper spectral inference methods to human physiological time-series data from ECG recordings for denoised frequency-domain heart rate variability analysis, in collaboration with Prof. Nicholas Schiff at Weill Cornell Medicine. [Python]

University of Peradeniya, Sri Lanka 2016 - 2017
Undergraduate and Postgraduate Researcher, Advisors: Dr. Janaka Wijayakulasooriya, Dr. Roshan Godaliyadda, and Dr. Parakrama Ekanayake

· Created a video database and developed a foreground estimation and trajectory tracking technique based on adaptive filtering, dynamic modeling, and affinity construction for trajectories.
· Introduced methods for learning patterns of activity using machine learning techniques including K-means and spectral clustering.
· Developed machine learning methods for motion pattern analysis in video surveillance, event detection, and anomaly identification.
· Mentored undergraduate students in applying the developed clustering and pattern analysis methods to hyperspectral imaging-based land cover mapping.

TEACHING EXPERIENCE

Rutgers University, New Brunswick 2026 - Present
Part-Time Lecturer, Department of Biomedical Engineering

· Instructor for BME 309: Devices and Systems, a core undergraduate course for junior-level biomedical engineering students.
· Independently design and deliver lectures covering signals, systems, circuits, and their applications to biomedical engineering.
· Develop instructional materials, homework assignments, quizzes, and examinations, and provide support through office hours.

University of Maryland, College Park 2017 - 2019
Graduate Teaching Assistant, Department of Electrical & Computer Engineering

· Supported instruction in undergraduate Engineering Probability and graduate Random Processes in Communication and Control.
· Teaching Assistant Training and Development Fellow: mentoring graduate teaching assistants and conducting workshops.

University of Peradeniya, Sri Lanka 2016 - 2017
Temporary Instructor, Department of Electrical & Electronic Engineering

· Conducted laboratory and tutorial sessions for multiple undergraduate courses in signal processing and electronics.

INDUSTRY EXPERIENCE

Trainee Engineer *October 2015 - January 2016*
Dialog Axiata PLC - Sri Lanka
· Analyzed communication network systems and contributed to system planning and optimization tasks
· Developed technical documentation, including a disaster recovery plan

Trainee Engineer *October 2014 - January 2015*
Ceylon Electricity Board - Sri Lanka
· Gained hands-on experience with power distribution systems, planning, and operations

SELECTED PUBLICATIONS AND PRESENTATIONS

A. Rupasinghe and J. W. Pillow, "Continuous Multinomial Logistic Regression for Neural Decoding," International Conference on Learning Representations (ICLR), 2026. [Presented]
A. Rupasinghe, A. S. Charles, and J. W. Pillow, "Continuous partitioning of neuronal variability," eLife, 2026.
P. Jendrichovsky, S. Khosravi, A. Rupasinghe, et al., "Patchy harmonic functional connectivity of the mouse auditory cortex," Proceedings of the National Academy of Sciences (PNAS) U.S.A. 122(27): e2510012122, 2025.
A. Rupasinghe, N. A. Francis, J. Liu, Z. Bowen, P. O. Kanold, and B. Babadi, "Direct Extraction of Signal and Noise Correlations from Two-Photon Calcium Imaging of Ensemble Neuronal Activity," eLife, 10:e68046, 2021.
A. Rupasinghe and B. Babadi, "Multitaper Analysis of Semi-Stationary Spectra From Multivariate Neuronal Spiking Observations," IEEE Transactions on Signal Processing, Vol. 68, pp. 4328-4396, 2020.
A. Rupasinghe, S. Mukherjee, and B. Babadi, "Adaptive Frequency-domain Granger Causal Inference from Neuronal Ensemble Data," Proceedings of the 54th Asilomar Conference on Signals, Systems, and Computers, 2020. [Presented]
A. Rupasinghe, et al., "Modes of Clustering for Motion Pattern Analysis in Video Surveillance," Proceedings of the 8th IEEE International Conference on Information and Automation for Sustainability, 2016. [Presented]

SELECTED HONORS AND SCHOLARSHIPS

Recipient of the Princeton Neuroscience Institute's QCN T32 Postdoctoral Fellowship Award	2024
Recipient of the SfN Trainee Professional Development Award	2024
Recipient of the Rising Stars in EECS Award	2022
Outstanding Teaching Assistant, Department of Electrical & Computer Engineering, University of Maryland	2018
Clark School of Engineering Distinguished Graduate Fellowship, University of Maryland	2017
Best Performance in Engineering, University of Peradeniya, Sri Lanka	2016